

Last login: Mon Jun 29 09:58:45 on ttys000
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch-200mb-2
psql (18.4)
Type "help" for help.

tpch-200mb-2=# SET ROLE alice;

EXPLAIN (ANALYZE, VERBOSE, BUFFERS)

```
select
  o_orderpriority,
  count(*) as order_count
from
  orders
where
  o_orderdate >= date '1994-01-01'
  and o_orderdate < date '1994-01-01' + interval '3' month
  and exists (
    select
      *
    from
      lineitem
    where
      l_orderkey = o_orderkey
      and l_commitdate < l_receiptdate
  )
group by
  o_orderpriority
order by
  o_orderpriority;
```

SET

WARNING: Log 1: Planner Input:

DETAIL:

```
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE true
:isReturn false
:cteList <>
:rtables (
  {RANGETBLENTY
:alias <>
:eref
  {ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 18182
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
  {ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs (
  {VAR
:varno 1
:varattno 6
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 51
}
)
:lateral false
:inFromCl false
:securityQuals <>
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
:relid 18182
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
:rteindex 1
}
)
}
:qual
{BOOLEXP
:boolop and
:args (
  {OPEXP
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 134
}
  {CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 154
}
}
}
```

o_orderdate >= DATE '1994-01-01'

```

:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
)
:location 146
}
{OEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varatype 1082
:varitypm 0
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 179
}
{OEXPR
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 198
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
{CONST
:consttype 1186
:consttypmod 196607
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 222
:constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
}
}
:location 211
}
)
:location 191
}
{SUBLINK
:sublinktype 0
:sublinkid 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
"l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
t")
}
}
:rtkind 0
:relid 18194
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromC1 true
:securityQuals (
{BOOLEXP
:boolop and
:args (
{SUBLINK
:sublinktype 0
:sublinkid 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname o
:colnames <>
}
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orders
tatus" "o_totalprice" "o_orderdate" "o_orderp
riority" "o_clerk" "o_shippriority" "o_commen
t")
}
}
}
}
}

```

o_orderdate < DATE '1994-01-01' + INTERVAL '3' MONTH

```

:rtkind 0
:relid 18182
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18182
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varatype 23
:varitypm 1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
)
{VAR
:varno 1
:varattno 1
:varatype 23
:varitypm 1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
)
:location -1
}
)
{SCALARARRAYOPEXP
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
{VAR
:varno 1
:varattno 6
:varitypm 1042
:varitypm 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
)
{ARRAYEXP
:array_typeid 1014
:array_collid 100
:element_typeid 1042
:elements (
{CONST
:consttype 1042
:constitypm 1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 12 [ 48 0 0 0 49 45 85 8
2 71 69 78 84 ]
}
{CONST
:consttype 1042
:constitypm 1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 50 45 72 7
3 71 72 ]
}
)
:multidims false
:list_start -1
:list_end -1
:location -1
}
)
:location -1
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
{TARGETENTRY
:expr
{CONST
:consttype 23
:constitypm 1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
}
)

```

l_orderkey = o_orderkey

o_orderpriority IN ('1-URGENT', '2-HIGH')

```

        :resno 1
        :resname ?column?
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
}
)
:override 0
:onConflict <>
:ReturningOldAlias <>
:ReturningNewAlias <>
:ReturningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTRY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_suppkey" "ps_avai
lty" "ps_supplycost" "ps_comment")
}
:rtekind 0
:relid 18163
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18163
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
):jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtable 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
)
}
:location -1
}
)
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (

```

ps_partkey = l_partkey

```

{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
}
:location -1
}
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 0 ]
}
}
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:location -1
}
}
)
}
)
:rtperminfos (
{RTPERMISSIONINFO
:relid 18194
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
23)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXPRESS
:boolop and
:args (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16

```

ps_suppkey =
l_suppkey

ps_availqty > 0

```

:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 402
  }
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 415
  }
)
:location 413
}
{OPEXPR
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 12
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 12
    :location 454
  }
  {VAR
    :varno 1
    :varattno 13
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 13
    :location 469
  }
)
:location 467
}
}
:location 450
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location 300
    }
  }
  :resno 1
  :resname l_orderkey
  :ressortgroupref 0
  :resorigtbl 18194
  :resorigcol 1
  :resjunk false
}
{TARGETENTRY
:expr
  {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location 300
  }
}
:resno 2
:resname l_partkey
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
  {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location 300
  }
}
:resno 3
:resname l_suppkey
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
  {VAR
    :varno 1

```

o_orderkey =
l_orderkey

l_commitdate <
l_receiptdate

```

:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 300
}
:resno 4
:resname 1_linenummer
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 300
}
:resno 5
:resname 1_quantity
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 300
}
:resno 6
:resname 1_extendedprice
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 300
}
:resno 7
:resname 1_discount
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 300
}
:resno 8
:resname 1_tax
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 300
}
:resno 9
:resname 1_returnflag
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 300
}
:resno 10
:resname 1_linestatus
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 10

```

```

:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 300
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 300
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 300
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location 300
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 1042
:vartypmod 14
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location 300
}
:resno 15
:resname l_shipmode
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 1043
:vartypmod 48
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location 300
}
:resno 16
:resname l_comment
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 16
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>

```

```

:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 244
}
)
:location 175
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 1
:varstype 1042
:varitypm 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname o_orderpriority
:ressortgroupref 1
:resorigtbl 18182
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 76
}
:resno 2
:resname order_count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: Log 2 : After pull up At starting of Path Generation:

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname o
:colnames <>
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:reloid 18182

```

```

:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18182
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
):jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varctype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
)
{PARAM
:paramkind 1
:paramid 0
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
){SCALARARRAYOPEXPR
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
{VAR
:varno 1
:varattno 6
:varctype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
){CONST
:consttype 1014
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0 ]
}
)
}
:location -1
}
)
):mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
)
)

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {SCALARARRAYOPEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args ({VAR :varno 1 :varattno 6 :var
type 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -
1 :constbyval false :constisnull false :location -1 :constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0 ]}) :location -1}
:is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1)
:incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies
<> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRIC
INFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :varctype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlev
elsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :is_pushed_down true :can_jo
in false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :
outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVAL
ENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :varctype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_
is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcoll
idsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}}
)
)
WARNING: is_parallel_safe: hazard_walker: Param paramid=0 in_safe_list=0 → reason=outer_reference
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {SCALARARRAYOPEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args ({VAR :varno 1 :varattno 6 :va
rtype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -1
:constbyval false :constisnull false :location -1 :constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0 ]}) :location -1}
:is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b
1) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies
<> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRIC
TINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :varctype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varle
velsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :is_pushed_down true :can_jo
in false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :
outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVAL
ENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :varctype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_
is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcoll
idsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}}
)
)

```

```

velsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_j
oin false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relids (b 1) :required_relids (b 1) :incompatible_relids (b
) :outer_relids (b 1) :left_relids (b 1) :right_relids (b 1) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVA
LENCEMEMBER :em_expr (VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b 1) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) :em_relids (b 1) :em
_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relids (b 1)}) :right_em (EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcol
lid 0 :location -1} :em_relids (b 1) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relids (b 1)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_buc
ketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} | parallel_safe = false | reason = PROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2: After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTRY
:alias
{ALIAS
:aliasname o
:colnames <>
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o
_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtkind 0
:reld 16182
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperinfos (
{RTEPERMISSIONINFO
:reld 16182
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 0 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{SCALARARRAYOPEXPR
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
{VAR
:varno 1
:varattno 6
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
){CONST
:consttype 1014
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0
0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0
]
}
)
:location -1
}
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>

```

```

:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {SCALARARRAYOPEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args ({VAR :varno 1 :varattno 6 :var
type 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -
1 :constbval false :constisnull false :location -1 :constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0 ]}) :location -1}
:is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1
) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies
<> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {SCALARARRAYOPEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args ({VAR :varno 1 :varattno 6 :va
rtype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen
-1 :constbval false :constisnull false :location -1 :constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0 ]}) :location -1}
:is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b
1) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies
<> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | paral
lel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:teletst <>
:rttable (
{RANGETBLENTRY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_suppley" "ps_availqty" "ps_supplycost" "p
s_comment")
}
:rtkind 0
:relid 18163
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rtperminfos (
{RTEPERMISSIONINFO
:relid 18163
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 2
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
}
:location -1
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0

```



```

        :colnames <>
    }
}
:eref
{
    (ALIAS
    :aliasname ps
    :colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "p
s_comment")
    )
}
:rtekind 0
:relid 18163
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18163
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
){CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
)

```

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 521 :opfuncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :

```

vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true
:constisnull false :location -1 :constvalue 4 [ 0 0 0 0 0 0 0 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volat
ile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_co
st :startup -1 :eval_cost_per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_m
cvtfreq -1 :right_mcvtfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: INPUT qual = {(RESTRICTINFO :clause {OEXPR :opno 521 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 3 :vartype 23
:vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval tru
e :constisnull false :location -1 :constvalue 4 [ 0 0 0 0 0 0 0 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_vola
tile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_c
ost :startup -1 :eval_cost_per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_m
cvtfreq -1 :right_mcvtfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = {(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
{VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = {(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
} {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1}) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = {(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
{VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = {(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
} {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1}) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {
  QUERY
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStmt <>
  :resultRelation 0
  :hasAggs true
  :hasWindowFuncs false
  :hasTargetSRFs false
  :hasSublinks true
  :hasDistinctOn false
  :hasRecursive false
  :hasModifyingCTE false
  :hasForUpdate false
  :hasRowSecurity true
  :hasGroupRTe true
  :isReturn false
  :rteList <>
  :rtable (
    {RANGETBENTRY
      :alias <>
      :eref
        {ALIAS
          :aliasname orders
          :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o
            _orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
        }
      :rtekind 0
      :relid 18182
      :inh false
      :relkind r
      :rellockmode 1
      :perminfoindex 1
      :tablesample <>
      :lateral false
      :inFromCl true
      :securityQuals <>
    }
    {RANGETBENTRY
      :alias <>
      :eref
        {ALIAS
          :aliasname *GROUP*
          :colnames ("o_orderpriority")
        }
      :rtekind 9
      :groupexprs (
        {VAR
          :varno 1
          :varattno 6
          :vartype 1042
          :vartypmod 19
          :varcollid 100
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 6
          :location 51
        }
      )
      :lateral false
      :inFromCl false
      :securityQuals <>
    }
    {RANGETBENTRY
      :alias <>
      :eref
        {ALIAS
          :aliasname lineitem
          :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_qua
            ntity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines
            tatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "
            l_shipmode" "l_comment")
        }
      :rtekind 0
      :relid 18194
      :inh false
      :relkind r
      :rellockmode 1
      :perminfoindex 2
      :tablesample <>
      :lateral false
      :inFromCl true
      :securityQuals ((
        {ALTERNATIVESUBPLAN
          :subplans (
            {SUBPLAN
              :subLinkType 0
              :testexpr <>
              :params <>
              :plan_id 1
              :plan_name SubPlan\ 1
              :firstColType 2278
              :firstColTypmod -1
              :firstColCollation 0
              :useHashTable false
              :unknownEqFalse true
              :parallel_safe false
              :setParam <>
              :parParam (i 0)
              :args (
                {VAR
                  :varno 3
                  :varattno 1
                  :vartype 23
                  :vartypmod -1
                  :varcollid 0
                  :varnullingrels (b)
                  :varlevelsup 0
                  :varreturningtype 0
                  :varnosyn 3
                  :varattnosyn 1
                  :location -1
                }
              )
              :startup_cost 0
              :per_call_cost 9772
            }
            {SUBPLAN
              :subLinkType 2
              :testexpr

```

```

{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
  {PARAM
:paramkind 1
:paramid 1
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
:paramids (i 1)
:plan_id 2
:plan_name SubPlan\ 2
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 9323.6
:per_call_cost 0.0025
}
)
}
{ALTERNATIVESUBPLAN
:subplans (
  {SUBPLAN
:subLinkType 0
:testexpr <>
:paramids <>
:plan_id 3
:plan_name SubPlan\ 3
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 2 3)
:args (
  {VAR
:varno 3
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 2
:location -1
}
  {VAR
:varno 3
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 3
:location -1
}
)
:startup_cost 0
:per_call_cost 6344
}
  {SUBPLAN
:subLinkType 2
:testexpr
{BOOLEXPR
:boolop and
:args (
  {OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
:varno 3
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 2
:location -1
}
    {PARAM
:paramkind 1
:paramid 4
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
  )
:location -1
}
  {OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
:varno 3
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0

```

```

        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 3
        :location -1
    }
    {PARAM
    :paramkind 1
    :paramid 5
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
    }
    }
    :location -1
    }
    )
    :location -1
    }
    :paramids (i 4 5)
    :plan_id 4
    :plan_name SubPlan\ 4
    :firstColType 23
    :firstColTypmod -1
    :firstColCollation 0
    :useHashTable true
    :unknownEqFalse true
    :parallel_safe true
    :setParam <>
    :parParam <>
    :args <>
    :startup_cost 5943.96
    :per_call_cost 0.005
    }
    )
    }
    ))
    }
    )
    :rteperminfos (
    {RTEPERMISSIONINFO
    :reliid 18182
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 12 13)
    :insertedCols (b)
    :updatedCols (b)
    }
    {RTEPERMISSIONINFO
    :reliid 18194
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
    :insertedCols (b)
    :updatedCols (b)
    }
    )
    )
    :jointree
    {FROMEXPR
    :fromlist (
    {JOINEXPR
    :jointype 4
    :isNatural false
    :larg
    {FROMEXPR
    :fromlist (
    {RANGETBLREF
    :rtindex 1
    }
    )
    }
    :quals (
    {OPEXPR
    :opno 1098
    :opfuncid 1090
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 1
    :varattno 5
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5
    :location 134
    }
    {CONST
    :consttype 1082
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 154
    :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
    }
    )
    }
    :location 146
    }
    {OPEXPR
    :opno 2345
    :opfuncid 2338
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 1
    :varattno 5
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5
    :location 179
    }
    {CONST
    :consttype 1114
    :consttypmod -1
    :constcollid 0
    :constlen 8
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
    }
    )
    }
    :location 191
    }
    )
    }
    :rarg
    {RANGETBLREF
    :rtindex 3

```

orders
SEMI JOIN lineitem

o_orderdate >= DATE
'1994-01-01'
AND
o_orderdate < DATE
'1994-04-01'

```

    }
:usingClause <>
:join_using_alias <>
:quals {
  {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args {
      {VAR
        :varno 3
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 1
        :location 402
      }
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location 415
      }
    }
  }
  :location 413
}
{OPEXPR
  :opno 1095
  :opfuncid 1007
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args {
    {VAR
      :varno 3
      :varattno 12
      :vartype 1082
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 3
      :varattnosyn 12
      :location 454
    }
    {VAR
      :varno 3
      :varattno 13
      :vartype 1082
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 3
      :varattnosyn 13
      :location 469
    }
  }
  :location 467
}
)
:alias <>
:rtindex 0
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList {
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 6
      :vartype 1042
      :vartypmod 19
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 6
      :location -1
    }
  }
  :resno 1
  :resname o_orderpriority
  :ressortgroupref 1
  :resorigtbl 10182
  :resorigcol 6
  :resjunk false
}
{TARGETENTRY
  :expr
  {AGGREF
    :aggfnoid 2803
    :aggtype 20
    :aggcollid 0
    :inputcollid 0
    :aggtranstype 0
    :aggargtypes <>
    :aggdirectargs <>
    :args <>
    :aggorder <>
    :aggdistinct <>
    :aggfilter <>
    :aggstar true
    :aggvariadic false
    :aggkind n
    :aggpresorted false
    :agglevelsup 0
    :aggssplit 0
    :aggno -1
    :aggtransno -1
    :location 76
  }
  :resno 2
  :resname order_count
  :ressortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>

```

lineitem.l_orderkey =
orders.o_orderkey

lineitem.l_commitdate <
lineitem.l_receiptdate

```
:groupClause (
  {SORTGROUPCLAUSE
  :tlesSortGroupRef 1
  :eqop 1054
  :sortop 1058
  :reverse_sort false
  :nulls_first false
  :hashable true
  }
)
:groupDistinct false
:groupingSets <>
:havingQual <>
:windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
  :tlesSortGroupRef 1
  :eqop 1054
  :sortop 1058
  :reverse_sort false
  :nulls_first false
  :hashable true
  }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 1098 :opfuncid 1098 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varratno 5 :vartype 10
82 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 1 :varratnosyn 5 :location -1) {CONST :consttype 1082 :consttypmod -1 :constcollid 0 :constlen 4 :constbyva
1 true :constisnull false :location -1 :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof t
rue :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_ser
ial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_buckets
ize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 2345 :opfuncid 2338 :opresulttype 16 :opretset false :opcollid 0 :inputcollid
0 :args ((VAR :varno 1 :varratno 5 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 1 :varratnosyn 5 :location -1) {CONST :consttype 1114 :cons
ttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant fal
se :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_reld
s (b 1) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hash
joinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 1098 :opfuncid 1098 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varratno 5 :vartype 1
082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 1 :varratnosyn 5 :location -1) {CONST :consttype 1082 :consttypmod -1 :constcollid 0 :constlen 4 :constbyv
al true :constisnull false :location -1 :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof
true :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_ser
ial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_buckets
ize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 2345 :opfuncid 2338 :opresulttype 16 :opretset false :opcollid 0 :inputcollid
0 :args ((VAR :varno 1 :varratno 5 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 1 :varratnosyn 5 :location -1) {CONST :consttype 1114 :cons
ttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant fa
lse :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_reld
s (b 1) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hash
joinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0} | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varratno 6 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 1 :varratnosyn 6 :location
-1) (VAR :varno 1 :varratno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 1 :varratnosyn 1 :location -1)
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varratno 6 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 1 :varratnosyn 6 :locatio
n -1) (VAR :varno 1 :varratno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 1 :varratnosyn 1 :location -1) | parallel_safe = true | reason =
PROPAGALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {ALTERNATIVESUBPLAN :subplans ((SUBPLAN :sublinkType 0 :testexpr <> :paramIds <> :plan_id 1 :plan_name SubPlan\ 1 :firstColType 2278 :firs
tColTypmod -1 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 0) :args ((VAR :varno 3 :varratno 1 :vartype 23 :vartypmod -1 :varcollid 0 :var
nullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 3 :varratnosyn 1 :location -1) :startup_cost 0 :per_call_cost 9772) (SUBPLAN :sublinkType 2 :testexpr {OPEXPR :opno 96 :opfuncid 65 :opresul
ttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 3 :varratno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 3 :varratnos
yn 1 :location -1) {PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :location -1} :paramIds (i 1) :plan_id 2 :plan_name SubPlan\ 2 :firstColType 23 :firstColTypm
od -1 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 9323.6 :per_call_cost 0.0025)}) :is_pushed_down true :can_join fal
se :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 3) :required_relds (b 3) :incompatible_relds (b) :outer_reld
s (b 1) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {ALTERNATIVESUBPLAN :
subplans ((SUBPLAN :sublinkType 0 :testexpr <> :paramIds <> :plan_id 3 :plan_name SubPlan\ 3 :firstColType 2278 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel
_safe false :setParam <> :parParam (i 2 3) :args ((VAR :varno 3 :varratno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 3 :varratnosyn 2 :locat
ion -1) (VAR :varno 3 :varratno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 3 :varratnosyn 3 :location -1) :startup_cost 0 :per_call_cost
6344) (SUBPLAN :sublinkType 2 :testexpr {BOOLEXPR :boolop and :args ((OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 3 :varratno 2 :vartype 2
3 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 3 :varratnosyn 2 :location -1) {PARAM :paramkind 1 :paramid 4 :paramtype 23 :paramtypmod -1 :paramcollid 0 :
location -1) :location -1) {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 3 :varratno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingr
els (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 3 :varratnosyn 3 :location -1) {PARAM :paramkind 1 :paramid 5 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :location -1) :paramIds (i 4 5) :plan_id 4 :plan_name SubPlan\ 4 :firstColType 23 :firstColTypmod -1 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args
<> :startup_cost 5943.96 :per_call_cost 0.005)}) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base
_rels 1 :clause_relds (b 3) :required_relds (b 3) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}
WARNING: max_parallel_hazard_walker: SubPlan plan_id=1 parallel_safe=0 : reason=correlated subplan
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {ALTERNATIVESUBPLAN :subplans ((SUBPLAN :sublinkType 0 :testexpr <> :paramIds <> :plan_id 1 :plan_name SubPlan\ 1 :firstColType 2278 :fir
stColTypmod -1 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 0) :args ((VAR :varno 3 :varratno 1 :vartype 23 :vartypmod -1 :varcollid 0 :va
rnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 3 :varratnosyn 1 :location -1) :startup_cost 0 :per_call_cost 9772) (SUBPLAN :sublinkType 2 :testexpr {OPEXPR :opno 96 :opfuncid 65 :opresul
ttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 3 :varratno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 3 :varratnos
yn 1 :location -1) {PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :location -1} :paramIds (i 1) :plan_id 2 :plan_name SubPlan\ 2 :firstColType 23 :firstColTyp
mod -1 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 9323.6 :per_call_cost 0.0025)}) :is_pushed_down true :can_join fa
lse :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 3) :required_relds (b 3) :incompatible_relds (b) :outer_reld
s (b 1) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :oute
r_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {ALTERNATIVESUBPLAN :
subplans ((SUBPLAN :sublinkType 0 :testexpr <> :paramIds <> :plan_id 3 :plan_name SubPlan\ 3 :firstColType 2278 :firstColCollation 0 :useHashTable false :unknownEqFalse true :paralle
l_safe false :setParam <> :parParam (i 2 3) :args ((VAR :varno 3 :varratno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 3 :varratnosyn 2 :locat
ion -1) (VAR :varno 3 :varratno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 3 :varratnosyn 3 :location -1) :startup_cost 0 :per_call_cost
6344) (SUBPLAN :sublinkType 2 :testexpr {BOOLEXPR :boolop and :args ((OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 3 :varratno 2 :vartype
23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 3 :varratnosyn 2 :location -1) {PARAM :paramkind 1 :paramid 4 :paramtype 23 :paramtypmod -1 :paramcollid 0
:location -1) :location -1) {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 3 :varratno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingr
els (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 3 :varratnosyn 3 :location -1) {PARAM :paramkind 1 :paramid 5 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :location -1) :paramIds (i 4 5) :plan_id 4 :plan_name SubPlan\ 4 :firstColType 23 :firstColTypmod -1 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args
<> :startup_cost 5943.96 :per_call_cost 0.005)}) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base
_rels 1 :clause_relds (b 3) :required_relds (b 3) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple
0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_ha
shoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 1095 :opfuncid 1087 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 3 :varratno 12 :vartype 108
2 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 3 :varratnosyn 12 :location -1) (VAR :varno 3 :varratno 13 :vartype 1082 :vartypmod -1 :varcollid 0 :varnul
lingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 3 :varratnosyn 13 :location -1) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakp
roof true :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relds (b 3) :required_relds (b 3) :incompatible_relds (b) :outer_relds (b) :left_relds (b 3) :right_relds (b 3) :orclause <> :r
info_serial 6 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_b
ucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varratno 6 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 1 :varratnosyn 6 :location
-1)
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varratno 6 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 1 :varratnosyn 6 :location
-1) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varratno 6 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 1 :varratnosyn 6 :location
-1) {AGGREG :aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilter <> :aggstar true :aggvariadic fals
e :aggkind n :aggpresorted false :agglevelsop 0 :aggsplit 0 :aggno 0 :aggtransno 0 :location -1) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varratno 6 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 1 :varratnosyn 6 :locatio
n -1) {AGGREG :aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilter <> :aggstar true :aggvariadic fals
e :aggkind n :aggpresorted false :agglevelsop 0 :aggsplit 0 :aggno 0 :aggtransno 0 :location -1) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varratno 6 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 1 :varratnosyn 6 :location
-1) {AGGREG :aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilter <> :aggstar true :aggvariadic fals
e :aggkind n :aggpresorted false :agglevelsop 0 :aggsplit 0 :aggno 0 :aggtransno 0 :location -1)
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varratno 6 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 1 :varratnosyn 6 :locatio
n -1) {AGGREG :aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilter <> :aggstar true :aggvariadic fal
se :aggkind n :aggpresorted false :agglevelsop 0 :aggsplit 0 :aggno 0 :aggtransno 0 :location -1) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varratno 6 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 1 :varratnosyn 6 :locatio
n -1)
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varratno 6 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsop 0 :varreturningtype 0 :varnosyn 1 :varratnosyn 6 :locatio
n -1) | parallel_safe = true | reason = PROPAGALLEL_SAFE
```

QUERY PLAN

```
GroupAggregate (cost=8439.49..19355974985.08 rows=5 width=24) (actual time=46814.233..347876.643 rows=2..0 loops=1)
  Output: orders.o_orderpriority, count(*)
  Group Key: orders.o_orderpriority
  Buffers: shared hit=5594 read=31120, temp read=4816208 written=5822
  -> Nested Loop Semi Join (cost=8439.49..19355974965.89 rows=3628 width=16) (actual time=516.356..347872.856 rows=4183.00 loops=1)
    Output: orders.o_orderpriority
    Join Filter: (orders.o_orderkey = lineitem.l_orderkey)
    Rows Removed by Join Filter: 2816363815
```

```

Buffers: shared hit=5594 read=31120, temp read=4816208 written=522
-> Gather Merge (cost=8439.49..9777.11 rows=11485 width=20) (actual time=92.867..108.338 rows=11352.00 loops=1)
  Output: orders.o_orderpriority, orders.o_orderkey
  Workers Planned: 2
  Workers Launched: 2
  Buffers: shared hit=2438 read=2908
-> Sort (cost=7439.47..7451.43 rows=4785 width=20) (actual time=85.666..86.981 rows=3784.00 loops=3)
  Output: orders.o_orderpriority, orders.o_orderkey
  Sort Key: orders.o_orderpriority
  Sort Method: quicksort Memory: 377kB
  Buffers: shared hit=2438 read=2908
  Worker 0: actual time=82.154..82.429 rows=3314.00 loops=1
    Sort Method: quicksort Memory: 226kB
    Buffers: shared hit=832 read=782
  Worker 1: actual time=82.296..82.643 rows=3322.00 loops=1
    Sort Method: quicksort Memory: 226kB
    Buffers: shared hit=772 read=804
-> Parallel Seq Scan on public.orders (cost=0.00..7147.00 rows=4785 width=20) (actual time=0.679..83.825 rows=3784.00 loops=3)
  Output: orders.o_orderpriority, orders.o_orderkey
  Filter: ((orders.o_orderdate >= '1994-01-01'::date) AND (orders.o_orderdate < '1994-04-01 00:00:00'::timestamp without time zone))
  Rows Removed by Filter: 96216
  Buffers: shared hit=2364 read=2908
  Worker 0: actual time=1.149..80.653 rows=3314.00 loops=1
    Buffers: shared hit=795 read=782
  Worker 1: actual time=0.548..80.638 rows=3322.00 loops=1
    Buffers: shared hit=735 read=804
-> Materialize (cost=0.00..19338738455.60 rows=99997 width=4) (actual time=0.019..16.828 rows=248094.43 loops=11352)
  Output: lineitem.l_orderkey
  Storage: Disk Maximum Storage: 4159kB
  Buffers: shared hit=3156 read=28212, temp read=4816208 written=522
-> Seq Scan on public.lineitem (cost=0.00..19338737955.61 rows=99997 width=4) (actual time=182.223..777.791 rows=304150.00 loops=1)
  Output: lineitem.l_orderkey
  Filter: ((lineitem.l_commdtdate < lineitem.l_receiptdate) AND (ANY ((lineitem.l_partkey = (hashed SubPlan 4).col1) AND (lineitem.l_suppkey = (hashed SubPlan 4).col2)))) AND (lin
eitem.l_orderkey = (hashed SubPlan 2).col1)))
  Rows Removed by Filter: 895819
  Buffers: shared hit=3156 read=28212
  SubPlan 4
    -> Seq Scan on public.partsupp ps (cost=0.00..5544.00 rows=159984 width=8) (actual time=0.364..64.241 rows=160000.00 loops=1)
      Output: ps.ps_partkey, ps.ps_suppkey
      Filter: (ps.ps_availqty > 0)
      Buffers: shared hit=562 read=2982
  SubPlan 2
    -> Seq Scan on public.orders o (cost=0.00..9022.00 rows=120640 width=4) (actual time=0.100..57.238 rows=120144.00 loops=1)
      Output: o.o_orderkey
      Filter: (o.o_orderpriority = ANY ('{1-URGENT,2-HIGH}'::bpchar[]))
      Rows Removed by Filter: 179856
      Buffers: shared hit=2592 read=2680
Planning:
  Buffers: shared hit=202 read=1 dirtied=2
Planning Time: 7.978 ms
Execution Time: 347688.620 ms
(58 rows)

tpch-200mb-2=>

```